ORIGINAL ARTICLE

The Application of Virtual Simulation Teaching in the Education of Pediatric Acute Intussusception

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In the field of medical education, traditional teaching methods of ten fall short of achieving the desired educational outcomes for surgical diseases. Virtual simulation teaching, which utilizes computer-generated 3D models and interactive simulations, provides learners with a safe, controllable, and repeatable learning environment. This study focuses on pediatric acute intussusception as the teaching content, exploring the effectiveness and role of virtual simulation teaching as a new method in the education of pediatric surgical diseases. (J Surg Ed 82:103536. © 2025 Published by Elsevier Inc. on behalf of Association of Program Directors in Surgery.)

KEY WORDS: acute intussusception, learning effect, traditional classroom teaching, virtual simulation teaching

COMPETENCIES: Patient Care, Medical Knowledge, Interpersonal and Communication Skills, Practice-Based Learning and Improvement

INTRODUCTION

In the field of medical education, traditional teaching methods often rely on theoretical lectures and limited practical opportunities. This approach is challenging to achieve ideal educational outcomes for certain surgical conditions that require a high degree of interactive and practical teaching.[1] Pediatric acute intussusception is a common pediatric surgical emergency. In the author's medical center, approximately 1000 such cases are treated annually. The diagnosis and treatment of this condition demand that physicians possess advanced professional knowledge and skills.[2] However, traditional classroom teaching in the context of this disease provides medical students with insufficient practical opportunities, thereby limiting their understanding of the disease and their mastery of diagnostic and treatment skills.

With the advancement of artificial intelligence (AI) and virtual simulation technology, a new teaching model has emerged. Virtual simulation teaching utilizes computer-generated 3-dimensional models and interactive simulations to provide learners with a safe, controllable, and repeatable learning environment.[3] In various surgical teaching scenarios, virtual simulation technology has been proven to effectively enhance medical students' learning interest, deepen their understanding of disease mechanisms, and improve their surgical skills proficiency.[4] Applying virtual simulation technology to the teaching of acute intussusception in children not only overcomes the limitations of traditional teaching models but also provides medical students with a learning experience that closely approximates reality. By simulating real clinical situations, learners can practice diagnosis and treatment processes in a risk-free environment, better preparing them to face actual medical challenges. Additionally, virtual simulation teaching can offer personalized learning paths and feedback based on learners' progress and needs, which contributes to improving the efficiency and effectiveness of the teaching.[5] The virtual simulation teaching of acute intussusception in children is also advanced, integrating the latest medical research findings and clinical practices to ensure the timeliness and accuracy of the teaching content. Through this teaching model, medical students can not only learn the latest medical knowledge and

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techniques but also cultivate their clinical thinking and decision-making abilities. Furthermore, virtual simulation teaching fosters interdisciplinary communication and collaboration, providing medical students with a broader learning perspective.[6]

This study will focus on acute intussusception in children as the teaching content and compare the learning outcomes of 2 groups of students, 1 using traditional classroom teaching and the other employing virtual simulation teaching. The research aims to explore the effectiveness and role of virtual simulation teaching as a new pedagogical approach in the teaching of pediatric surgical diseases.

METHODS

To thoroughly investigate the application effects of virtual simulation teaching in the education of pediatric acute intussusception, this study has designed a randomized controlled trial. The objective of this trial is to evaluate the efficacy of 2 teaching methods—traditional instruction and virtual simulation instruction—in improving students' standardized test scores, practical application skills, learning motivation and attitudes, as well as the quality of instructional feedback.

Study Design

(1) Sample Selection: A total of 212 medical students were selected, divided into 2 equal-sized groups of 106 each, ensuring comparability in gender, age, and baseline medical knowledge levels.
(2) Random Allocation: Students were randomly assigned to 2 groups using a random sampling method, with 1 group receiving traditional teaching methods and the other group receiving virtual simulation teaching methods.
(3) Instructor Selection: Both groups, utilizing different teaching methods, were instructed by the same teacher.

Teaching Methods

The teaching content for both the traditional teaching group and the virtual simulation group was the same, focusing on pediatric acute intussusception.

(1) Traditional Teaching Group: Utilizes traditional lecture delivery and slide presentation. The teaching content for the traditional teaching group includes: the definition, etiology, clinical manifestations, physical examination, ancillary examinations, diagnosis and differential diagnosis, and treatment methods of pediatric acute intussusception.
(2) Virtual Simulation Teaching Group: Employs virtual simulation technology to provide a 3-dimensional simulated environment, enabling students to engage in interactive learning and diagnostic simulation. The teaching content for the virtual simulation group includes: Trainees log in using their personal accounts and passwords to participate in the training/assessment of clinical diagnostic and treatment skills. The system supports both self - study and assessment modes. Trainees can understand their level of clinical diagnostic and treatment skills through the assessment mode and improve their clinical diagnostic thinking skills using the self—study mode. The learning steps include studying theoretical knowledge related to pediatric intussusception under the guidance of teachers, and then trainees can perform operations such as taking medical history, physical examination, ancillary examinations, diagnosis, and treatment on virtual patients. The system supports the evolution of the disease process. After trainees provide treatment to virtual patients, they can see the progression of the patient's condition and adjust the treatment plan according to the changes in the condition until the treatment is completed.

Assessment Indicators

(1) Standardized Performance and Practical Application Ability: Evaluates students' grasp of theoretical knowledge on acute intussusception in children through postclass theoretical assessments and the completion time and accuracy of simulated air enema reduction operations.
(2) Learning Motivation and Attitude: Assesses students' learning interest (categorized as very interested, interested, not interested), learning experience (categorized as very satisfied, satisfied, dissatisfied), and self-efficacy through questionnaire surveys.
(3) Teaching Feedback: Collects evaluations from instructors on students (categorized as excellent, good, pass, fail) and students' satisfaction with teaching methods (categorized as very satisfied, satisfied, dissatisfied).

Data Collection and Analysis

The aforementioned assessment indicators are collected after the completion of teaching and evaluation. For standardized performance and practical application ability

indicators that do not meet the normality test for distribution, the Wilcoxon test is employed. For learning motivation and attitude, as well as teaching feedback indicators, the Kruskal-Wallis H test for ordinal variable difference is utilized.

RESULTS

Standardized Performance and Practical Application Ability

Firstly, we compared the differences in standardized performance and practical application ability between 2 groups of students using different learning methods. The observed indicators were quantifiable metrics: theoretical examination scores, completion time of simulated operation assessments, and completion degree of simulated operation assessments. The following Table 1 presents the overall statistical situation. It can be observed that, in terms of mean values and ranges, the virtual simulation teaching group outperformed the traditional teaching group.

The specific statistical analysis and visualization results are as follows:

For the data of the 3 observed indicators, the Kolmogorov-Smirnov test was applied, yielding a p-value < 0.05, thus rejecting the normality assumption. Consequently, the Wilcoxon Rank Sum Test was employed. The statistical results of the 3 datasets from the virtual simulation teaching group were all superior to those of the traditional teaching group, with a p-value < 0.05.

Learning Motivation and Learning Attitude

For the aforementioned indicators, we separately utilized the observation of learning interest (1. Highly Interested, 2. Interested, 3. Not Interested) and learning experience (1. Highly Satisfied, 2. Satisfied, 3. Not satisfied) to conduct statistical analysis. The results demonstrated that students in the virtual simulation teaching group exhibited significantly superior learning motivation and attitude compared to those in the traditional teaching group, with no students reporting disinterest or dissatisfaction.

In the statistical analysis of the aforementioned indicators, since learning interest is an ordinal variable (there is an order of superiority and inferiority between different categories), the common Chi-square Test is not applicable. Instead, the Kruskal-Wallis H rank sum test, which is more suitable for testing differences in ordinal variables, is used. The visualization of the statistical analysis is shown in the following figure.

TABLE 1. Statistical Analysis of Objective Teaching Performance: Virtual Simulation Teaching Group vs. Traditional Teaching Group

Group	(Theoretical Assessment)			Practical Simulation Test Completion Time (Minutes)			Practical Simulation Test Proficiency Score (100%)		
	Median (IQR)	Mean (SD)	Range	Median (IQR)	Mean (SD)	Range	Median (IQR)	Mean (SD)	Range
traditional teaching group, N = 106	79.00 (75.00, 84.00)	78.93 (6.34)	63.00–91.00	5.47 (5.23, 6.22)	5.65 (0.63)	4.25–7.12	80.00 (75.00, 85.00)	78.77 (8.22)	60.00–95.00
virtual simulation instructional teaching group, N = 106	84.00 (80.00, 87.00)	83.19 (5.42)	65.00–94.00	5.22 (5.08, 5.42	5.29 (0.51)	4.30–6.55	85.00 (80.00, 90.00)	84.10 (7.43)	65.00–100.00

Teaching Feedback

Collected were the evaluations of students by teachers from both the Virtual Simulation Teaching Group and the Traditional Teaching Group, categorized into Outstanding, Satisfactory, Passing, and Below Standard, as well as the students' satisfaction feedback on the teaching methods, categorized into Extremely Pleased, Pleased, and Displeased. The statistical results indicate that the Virtual Simulation Teaching Group significantly outperforms the Traditional Teaching Group in terms of teacher-student interaction evaluations, with a much higher proportion of Excellent and Very Satisfied ratings.

The statistical analysis of the aforementioned observation indicators, which include teachers' evaluations of students and students' satisfaction with teaching methods, also employs the Kruskal-Wallis H test, a suitable method for testing differences in ordinal variables. The visual representation of this statistical analysis is shown in the following figure.

DISCUSSION

Traditional classroom teaching methods in surgery have a long-standing history, tracing back to the early stages of medical education. Within this pedagogical framework, the transmission of knowledge predominantly relies on the teachers' lectures and demonstrations. Students learn and master surgical skills through listening, observing, and practical application.[7] Pediatric acute intussusception is 1 of the most common acute abdominal emergencies in pediatric surgery. Historically, the primary diagnostic and therapeutic approach for this condition has been the reduction of the intussuscepted bowel using air or water enema, a method that has predominantly been taught through classroom instruction.[8] This teaching method, which involves lectures with PowerPoint presentations and demonstrations using teaching aids, has long been the mainstay in medical education. It has played a crucial role in imparting fundamental theories and clinical procedural skills.

However, with the rapid development of information technology, traditional surgical classroom teaching methods have gradually revealed some disadvantages and insufficiencies. Firstly, this teaching model often lacks interactivity and personalization, making it difficult to meet the learning needs of different students.[9] Secondly, traditional classrooms struggle to provide a sufficient number of practical opportunities, which can result in students not gaining adequate hands-on experience. Furthermore, the pace at which traditional teaching methods update knowledge is relatively slow, potentially failing to reflect the latest medical research findings and technological advancements in a timely manner.[10] In the information age, both students and educators expect more flexible, efficient, and modern teaching methods to adapt to the rapidly changing field of medicine. Consequently, traditional surgical classroom teaching methods are facing pressure to reform and innovate.

In recent years, the emergence of AI virtual teaching has increasingly highlighted its importance in surgical education. By simulating real surgical environments, it provides learners with a risk-free practice platform.[4,11] As early as 2014, the Stanford University School of Medicine reported the creation of highly realistic clinical scenarios through AI technology, allowing medical students to practice surgical operations in a virtual environment, thereby deepening their understanding of anatomical structures and surgical techniques.[11] In 2019, Harvard Medical School utilized Augmented Reality (AR) technology, enabling students to superimpose virtual anatomical structures onto actual human body models; this interactive learning method helps students to more intuitively understand complex anatomical relationships.[12] In 2020, the Johns Hopkins University School of Medicine developed an AI-based learning system that provides personalized educational content and feedback according to students' learning progress and comprehension abilities.[13,14] Furthermore, AI-assisted teaching systems can offer customized learning pathways and immediate feedback, enhancing the efficiency and quality of learning.[15,16] Compared to traditional classroom teaching, AI virtual teaching has significant advantages. It breaks through the limitations of time and space, allowing students to learn anytime and anywhere.[17] At the same time, virtual teaching can provide customized training modules based on students' learning progress and performance, achieving personalized education.[18,19] The real-time feedback mechanism of the AI system also greatly facilitates the rapid improvement of students' skills. Compared to traditional teaching models, AI virtual teaching shows tremendous potential in increasing learning motivation, optimizing the allocation of educational resources, and enhancing clinical skill training.[20,21]

Our study focuses on pediatric acute intussusception, a common acute abdominal emergency, and compares the teaching effectiveness between 2 groups of students using traditional classroom instruction and virtual simulation teaching methods. The observed indicators included objective metrics such as exam scores and the time and completeness of simulated treatment, as well as subjective metrics like motivation and attitude towards learning, and

feedback on teaching. The results showed that the virtual simulation teaching group significantly outperformed the control group in both the number of high achievers and the average scores of the students. Moreover, due to the unparalleled visual advantages of virtual simulation teaching, students in this group had a significant advantage in the simulated operation assessment. In addition, virtual simulation teaching was also markedly superior to traditional teaching in terms of teacher-student interaction and stimulating students' interest in learning, leading to a noticeable increase in satisfaction with the teaching and learning process for both teachers and students. Therefore, we believe that virtual simulation teaching has significant advantages over traditional classroom teaching in pediatric surgery education.

The application of AI virtual simulation teaching in the field of surgery is leading the innovation of medical education. Through highly realistic simulated environments, AI technology provides a safe and controllable learning platform for surgeons, especially for teaching diseases with high incidence rates and treatment requiring visual training, such as pediatric acute intussusception. It can simulate various situations in actual surgery, allowing doctors to hone their skills in a risk-free environment. Furthermore, AI-assisted teaching systems can provide

FIGURE 1. Comparison of objective teaching performance between virtual simulation teaching group and traditional teaching group (A. Statistical comparison of theoretical examination scores, B. Statistical comparison of completion time in simulation operation examination, C. Statistical comparison of completion degree in simulation operation examination).

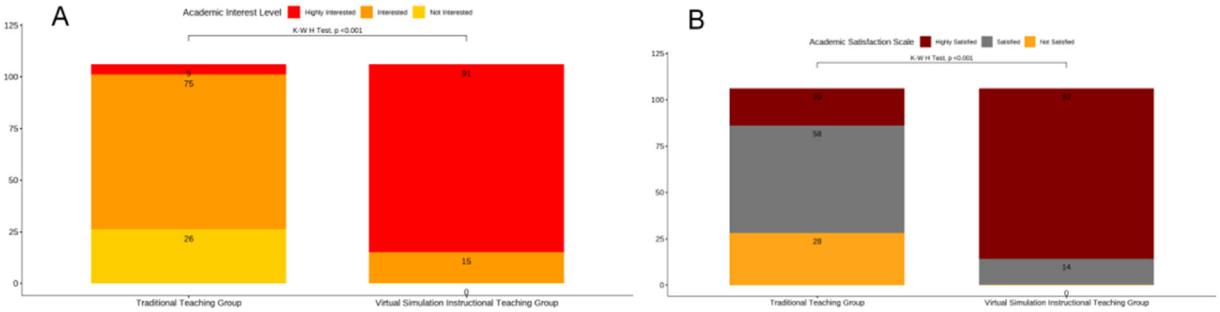

FIGURE 2. Comparison of learning motivation and attitudes between virtual simulation teaching group and Traditional teaching group (A. Statistical comparison of learning interest, B. Statistical comparison of learning experience).

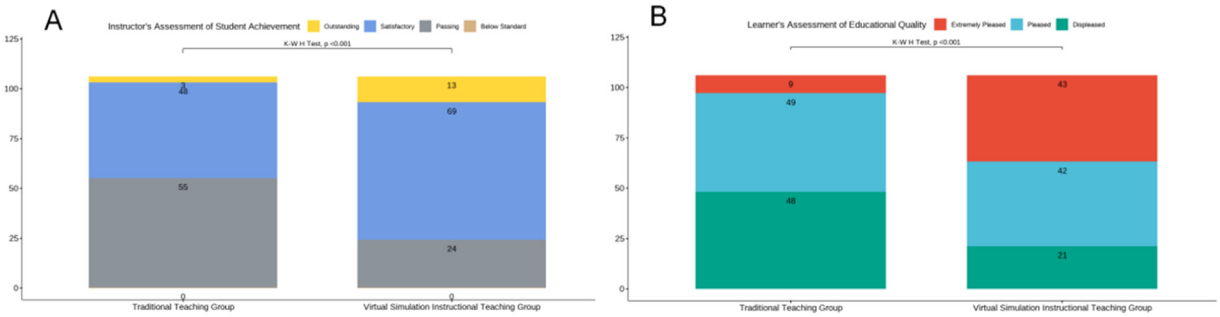

FIGURE 3. Comparison of teaching feedback between virtual simulation teaching group and traditional teaching group (A. Comparison of teachers' evaluations of students, B. Comparison of students' satisfaction with teaching methods).

TABLE 2. Statistics on Learning Motivation and Attitude of Virtual Simulation Teaching Group and Traditional Teaching Group

Characteristic	Overall, N = 212	Traditional Teaching Group, N = 106	Virtual Simulation Instructional Teaching Group, N = 106
Academic interest Level, n (%)			
Highly interested	96 (45.3%)	5 (4.72%)	91 (85.8%)
Interested	90 (42.5%)	75 (70.8%)	15 (14.2%)
Not interested	26 (12.3%)	26 (24.5%)	0 (0%)
Academic satisfaction scale, n (%)			
Highly satisfied	112 (52.8%)	20 (18.9%)	92 (86.8%)
Satisfied	72 (34.0%)	58 (54.7%)	14 (13.2%)
Not satisfied	28 (13.2%)	28 (26.4%)	0 (0%)

TABLE 3. Statistical Analysis of Teaching Feedback for Virtual Simulation Teaching Group and Traditional Teaching Group

Characteristic	Overall, N = 212	Traditional Teaching Group, N = 106	Virtual Simulation Instructional Teaching Group, N = 106
Instructor's assessment of student achievement, n (%)			
Outstanding	16 (7.55%)	3 (2.83%)	13 (12.3%)
Satisfactory	117 (55.2%)	48 (45.3%)	69 (65.1%)
Passing	79 (37.3%)	55 (51.9%)	24 (22.6%)
Below standard	0 (0%)	0 (0%)	0 (0%)
Learner's assessment of educational quality, n (%)			
Extremely pleased	52 (24.5%)	9 (8.49%)	43 (40.6%)
Pleased	91 (42.9%)	49 (46.2%)	42 (39.6%)
Displeased	69 (32.5%)	48 (45.3%)	21 (19.8%)

personalized learning pathways and immediate feedback, thereby enhancing teaching efficiency and the quality of learning. With the continuous advancement of technology, AI virtual teaching is expected to become an important pillar of future surgical education, providing strong support for the training of the next generation of surgeons (Figs. 1-3 and Tables 1-3).

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